

Measurement Errors in “Check-All-That-Apply” Questions within Self-Administered
Web and Paper Surveys of Michigan Local Government Officials

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Abstract

The study delves into measurement errors within the response behaviors of Michigan's local political leaders, particularly regarding their handling of "check-all-that-apply" questions. The local political leaders are an underexplored cohort influenced by public stereotypes and motivated by their reputations for career success. Their survey response behaviors are expected to be distinct from the general public with them being easily identifiable, their unique relationships with research institutions, and disruptive events being inclusive of their services. Utilizing 13 survey waves spanning 13 years from the Michigan Public Policy Survey (MPPS) data sets, comprising 58 qualified questions, 510,538 response observations, and approximately 2,700 respondents per survey wave, the study employs logistic regression models to scrutinize hypotheses derived from existing literature. For example, how well do primacy effects apply to local political leader respondents? While limitations curtailed deeper exploration, the study concludes with insights from quantitative analyses addressing concerns about primacy effects regarding survey item sequence and the incorporation of "None" and "Don't Know" options.

Keywords: longitudinal survey, self-administered survey, "check-all-that-apply" question, measurement errors, logistic regression analysis, local political officials

Motivations

Previous research has predominantly delved into understanding the distinctions in response behaviors between forced-choice questions and check-all-that-apply questions (Raisinski et al., 1994; Callegaro et al., 2015; Nicolaas et al., 2015). The emphases of those check-all-that-apply studies have been on variations across questionnaire modes or comparisons with yes-no questions with the same survey mode applied. Factors influencing marks of each option, such as the length of choice descriptions, the number of options, their order, and respondents' comprehension of questions, have been identified as crucial. These factors may introduce measurement errors and bias concerns and hurt data quality. However, a notable gap exists in examining the nuanced response behaviors, particularly among local political leaders. Limited attention has been given to understanding how local politics, assumed to be public service enthusiasts, differ from the general public.

The check-all-that-apply approach, also known as mark-all-that-apply, represents a distinctive type of multiple-choice survey (Dooley et al., 2010; Roßmann et al., 2018). In this method, respondents are presented with a list of answer options and are required to select the ones they find relevant to the product. This approach is characterized by minimal instructions, ease of execution, and quick completion. Existing literature often hypothesizes that forced-choice questions, when compared to check-all-that-apply questions, align more closely with deeper cognitive processing. In addition, the check-all-that-apply format might encourage a satisficing response strategy.

Why focus on local political officials?

The significance and potential contributions of studying a relatively small and less studied population group, such as local political officials, lie in their unique characteristics, which may be influenced by public stereotypes (Krause et al., 2023).

Firstly, local political officials are easily identifiable, and their success in their positions heavily depends on their reputations. Taking Michigan State in the U.S. as an example, there are typically two pathways to becoming a local political leader: by appointment or by winning an election. In either case, their reputations play a crucial role in successfully completing their tenures.

Considering the influence held by media and prestigious research institutions like the University of Michigan, local officials are more inclined to collaborate with researchers to gain insights about their service populations or to enhance their reputation. Additionally, since local political officials are a small and easily identifiable population, refusing to cooperate may have negative repercussions on their political careers. An illustration of the potentially different response patterns can be observed in the high average response rate across over 20 waves of survey investigation of the Michigan Public Policy Survey (MPPS), the data sets used for analytics in this study, at 71.38%, compared to other surveys targeting the general public, such as the National Health Interview Survey (NHIS) and the National Health and Nutrition Examination Survey (NHANES).

Secondly, local officials usually have unique partnerships with advocates and community organizations, which also maintain strong ties with institutions mentioned in the previous paragraph. The Michigan Public Policy Survey (MPPS) leverages its institution's close relationships with local community organizations to assist with survey promotions and boost even higher response rates. Maintaining positive relationships with these partners is also crucial for local political officials.

Lastly, unlike the general public, whose schedules may vary, local officials share a common determination to navigate through their diverse terms of service and manage disruptive events like elections. Additionally, some officials may also have local cultural events, such as deer hunting season, on their calendars. These events, though regular and possibly regional, hold undeniable significance for both the officials and the residents within their respective jurisdictions. These shared experiences and distinctive characteristics underscore the importance of studies that center on them. While not the primary focus of the MPPS, this study expands upon the original topics of interest of the MPPS (e.g., fiscal, budgetary, operational policy, etc.) to encompass a more methodological research project, leveraging MPPS data sets.

Expanding on these considerations, this research embarks on an innovative exploration into the response patterns specific to check-all-that-apply items when administered to local official elites participating as survey respondents. To clarify, the study does not set out to scrutinize the validity of check-all-that-apply responses due to inherent challenges in acquiring the “true values” within current data constraints (i.e., this study is unable to collect new data

from either methodology for validating current data). Instead, its primary objective is to present readers with the observed response patterns and assess the potential reliability by diagnosing measurement errors in the question format when delivered to local official elites.

Given constraints on initiating experiments, the research involves a comparative analysis across multiple waves of similar survey studies. As a non-experimental study, the research examines survey questions posed to political officials over the years. Surveys are conducted primarily through the web, considering challenges faced by political officials, such as over-surveying perceptions, increased distrust towards external entities, scheduling constraints, and limitations related to incentives. The study acknowledges “external entities” as individuals or organizations lacking partnerships with respondents’ working organizations. The expected conclusions of this study will emphasize response behavioral patterns and data quality improvement in survey design geared toward local leaders. If resources permit, comparisons between non-local leaders and local leaders’ responses will also be explored.

Literature Review

“Check all that apply” questions come in multiple names in current literature. “Mark all,” “multiple response format,” and “select all” are just a few instances (Callegaro et al., 2015). It also comes in two types: (1) single lists, where each option in rows is associated with one checkbox and is independent of other checkboxes; and (2) grids, where one response option in a row has at least two checkboxes and requires more than one-time considerations to complete. This type of question is among all a top commonly used question formats in self-administered survey modes (Safir, 2008). It features its design efficiency because multiple selections are allowed among a list of options within a survey item as long as they are considered applicable to the respondents (Smyth et al., 2006). Kunz and Fuchs (2018) also mention its flexibility regarding the unconstrained number of response alternatives that can be selected.

The check-all-that-apply question is specifically compatible with self-administered web or mail surveys because it is available in the HTML-box web design feature. On the other hand, it is hardly feasible to telephone survey modes and is usually transformed into a forced yes-no question because it can be awkward and challenging for the interviewers to request that interviewees memorize all response options read before responding (Smyth et al., 2006). As initiating web surveys becomes even more accessible and popular in multiple fields, check-all-that-apply questions are also more frequently seen in surveys.

Current literature has proposed several benefits of the use of the check-all-that-apply question format. Firstly, Roßmann et al. (2018) elaborated that organizing multiple items in a condensed matrix format leads to quicker responses, enhances ease of use, and increases respondent enjoyment due to the reduced perceived length of the questionnaire, enhanced question context coherence, and minimal instructions needed (Dooley et al., 2010). Kunz and Fuchs (2018) further emphasized its convenience for researchers to capture responses to a question. Aligned with respondents’ intuitions, check-all-that-apply questions have the advantage of allowing for slightly less cognitive processing compared to scaling questions

regarding product description sensory tests (Dooley et al., 2010). From a “user friendliness” perspective, evident in its ability to reduce response burden (Raisinski et al., 1994), its enduring and widespread usage is apparent.

Nonetheless, associated with all the benefits the check-all-that-apply question format provides, there are concerns about its use in data collection and analyses. Callegaro et al. (2015) highlighted from several prior studies that even though the implication of a blank checkbox is “no” (Lazo et al., 2016), there are four possible interpretations: (1) the response option does not apply, (2) the respondent has missed this entry, (3) the respondent is uncertain about the answer, and (4) the respondent opts-out from responding to the alternative. Unlike forced yes-no questions where three implications, yes, no, and item-nonresponse, are each mapped with a specific, corresponding endorsement type, this complexity drastically increases the burden of researchers in data analysis.

In addition, traditional survey measurement errors also exist in the check-all-that-apply question type. Smyth et al. (2006) found that primacy effects appear to be in the play for respondents who answered the question under the mean completion time. Roßmann et al. (2018) also pointed out that respondents may use satisficing as a response strategy (i.e., skim over or entirely bypass the cognitive processes of understanding, recalling information, and making judgments) due to the unrestricted amount a respondent can select and the perceived complexity with thus much information listed at once.

Hypotheses

To undertake rigorous quantitative research, this study formulates several hypotheses based on existing literature mentioned in the Literature Review section prior to developing the analysis plan.

For check-all-that-apply questions in lists:

1. The probability of each term being marked decreases with the length of the number of options of check-all-that-apply questions.
2. Primacy effects suggest that terms listed early have a higher chance of being selected than those listed later.
3. Longer descriptions in the initial options of check-all-that-apply questions lead to a reduced likelihood of each option being chosen.
4. The “Other” option, requiring a descriptive response, has the lowest probability of being selected compared to options without separate written responses.
5. The likelihood of selecting “None” or “Don’t know” is higher, if presented (Luskin and Bullock, 2011; Weisberg, 2018).

Data Source

This study draws upon datasets sourced from the U-M Michigan Public Policy Survey (MPPS), a comprehensive initiative that has systematically surveyed local officials in all general-purpose jurisdictions across Michigan since 2009. These datasets encompass limited publicly available data from the inception of the study until the year 2023. The

MPPS has maintained a consistent annual schedule of one to two surveys, each addressing different topics relevant to current local leaders. These surveys are distributed via email, with a typical field period of approximately 60-70 days (Fitzpatrick et al., 2022).

In the event of nonresponses, the study employs robust follow-up strategies, combining email reminders, phone calls, and hardcopy mailings in different orders, frequencies, and sometimes jurisdiction types, if the necessary information is known. The survey responses, available to the public in distinct files for each wave, present an amount of information on the perspectives and experiences of local officials. However, the publicly available data segments responses into separate files, making cross-analyzing survey errors challenging for external researchers not affiliated with the survey team. Consequently, the datasets used for analysis span from 2009 to 2021.

During the period surveyed from 2009 to 2021, this study focuses on a specific type of “check-all-that-apply” question, employing a single-item design where each item is treated independently for marking. There are 58 questions presented as lists within the specified survey waves. These list questions encompass a range of 4-25 alternative options.

Minimum	1st Quantile	Median	Mean	3rd Quantile	Maximum
4.00	7.00	10.00	10.94	14.00	25.00

Table 1. Descriptive Statistics of the Number of Alternative Options in the 58 Qualified Questions

Another format involves grids, where each item is accompanied by multiple boxes (columns) for selection. However, as only 2 grid questions are included in the waves and lack the statistical power for inference, they have been excluded from the analysis.

Given the original emphasis on public policy analysis, the questions predominantly involve factual and objective inquiries but also include attitudinal or subjective ones. This rich and diverse dataset serves as a valuable foundation for subsequent analyses, offering insights into the nuanced perspectives of local leaders over the years.

Dependent Variable

Among hundreds of variables from each wave, this study selected a binary variable to examine the outcome of interest: the probability of checkbox selection, labeled “response” in the analysis. Initially containing “N/A” values due to skipping certain questions as per instructions, this binary variable was refined to include only “marked” (1) and “non-marked” (0) values for analysis.

Independent Variable

Reiterating the hypotheses, the independent variables consist of various factors characterizing the attributes of the prompts. All of the 8 independent variables are manually encoded by myself. A codebook detailing all main effects to be included in the initial

consideration is outlined in Table 2.

Variable Name	Data Type	Explanation	Categorical Values
nu_character_prompt	numeric	Number of characters in the prompt excluding spaces and without instructions such as those in italics, but includes words in parentheses if not instruction	
professional_terms	binary	Indicator for the presence of supplemental explanations or examples in parentheses	0 = No; 1 = Yes
nu_option	numeric	Number of options within the question where the checkbox is located	
order_prompt	numeric	Sequence of the prompt within the question	
order_survey	numeric	Sequence of the question containing the prompt in the survey	
other	binary	Indicates if the prompt for this checkbox includes an “other” option	0 = No; 1 = Yes
none	binary	Indicates if the prompt for this checkbox includes a “None” option	0 = No; 1 = Yes
dk	binary	Indicates if the prompt for this checkbox includes a “Don’t Know” option	0 = No; 1 = Yes

Table 2. Codebook of Independent Variables in the Initial Model

For easier reading, the independent variables listed in Table 2 will be referred to as “# of characters in prompt,” “professional terms,” “# of options,” “order in question,” “order in survey,” “other,” “none,” and “don’t know” hereby in tables and equations, respectively. Interaction terms will be incorporated into the analysis plan as needed, with their necessity examined in the Analytics Approaches section.

Methodology

To evaluate the hypotheses, this study employs the logistic regression analysis methodology for the single list check-all-that-apply questions. This model analyzes response patterns and measurement errors in check-all-that-apply items administered to Michigan local political leaders. The results encompass estimated coefficients as the log odds, estimated standard errors across the sample, and statistical significance z-tests for each predictor in response to the hypotheses.

The equation for the single-list check-all-that-apply questions is specified mathematically as Equation 1. It will be modified, seeking an optimized model, after the necessity of including interaction terms and several examinations have been explored in the Analytics Approaches section.

$$\begin{aligned} \text{logit}(\text{Pr}(\text{possibility of marked}_i = 1)) = & \beta_0 + \beta_1 \# \text{ of characters in prompt}_i \\ & + \beta_2 \text{professional terms}_i + \beta_3 \# \text{ of options}_i + \beta_4 \text{order in question}_i \\ & + \beta_5 \text{order in survey}_i + \beta_6 \text{other}_i + \beta_7 \text{none}_i + \beta_8 \text{don't know}_i, \\ & \text{where } i \text{ indicates each item/option/term} \end{aligned}$$

Equation 1. Initial Logistic Regression Model

Analytics Approaches

Data Preprocessing

Initially, we conducted data cleaning by eliminating partially completed cases ($\text{idcompleted} = 0$) and extracting response data from “check-all-that-apply” questions. After filtering out incomplete responses, there are approximately 1500 respondents in each wave. Each completed case is assigned a unique identification number (respondent_id) in the raw data. However, these IDs do not consistently differentiate between different respondents across waves, precluding the consideration of both the mixed-modeling and marginal effect modeling approaches.

Some responses to earlier questions allowed respondents to skip subsequent questions, resulting in the presence of “NAs” in the data. Another round of data cleaning, removing the “NAs,” resulted in 510,538 observations. The number of distinct respondents is unfortunately unidentifiable.

Following the data cleaning, we employed the “gather” function to transform the data from long to wide format, ensuring that each row represents only one result from one checkbox. Subsequently, we manually crafted all independent variables introduced in the Data Source section, resulting in a dataset containing 9 variables.

Exploratory Data Analysis

In the preliminary data analysis, we performed a correlation analysis computed with the Spearman rank correlation coefficients between predictors to determine which interaction terms

should be included. With a combination of continuous and categorical variables, Figure 1 displays the heatmap representing the results of the correlation analysis.

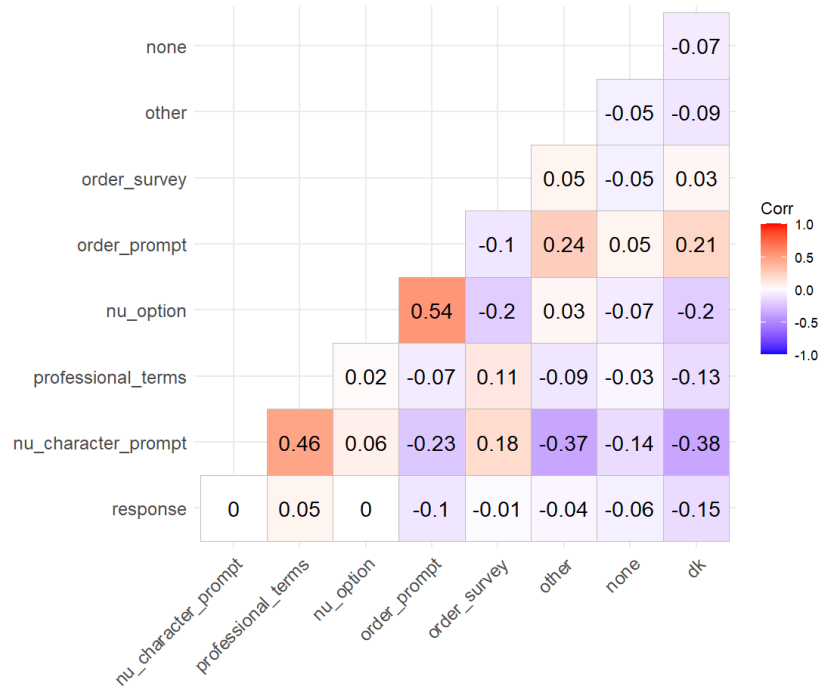


Figure 1. Heatmap of Correlation Analysis for Key Variables

Figure 1 indicates predominantly weak relationships among most variable pairs. However, moderate correlations are observed in 4 pairs of independent variables, including the number of characters in the prompt and the presence of supplemental explanations in parentheses, the number of options within the question and the sequence of the prompt within the question, the number of characters in the prompt and if the prompt indicates “other,” and the number of characters in the prompt and if the prompt indicates “don’t know.” Given correlation scores ranging between 0.3 and 0.7, interaction terms for these pairs are included in the initial full logistic regression model. However, through further exploration, the interaction between the number of characters in the prompt and if the prompt indicates “don’t know” is excluded. This is because this predictor perfectly predicts the probability of a checkbox being marked across our sample, leading to an infinite coefficient estimate.

The equation now encompasses 8 main effects and 3 interactions and is modified as Equation 2.

$$\begin{aligned}
 \text{logit}(\text{Pr}(\text{possibility of marked}_i = 1)) = & \beta_0 + \beta_1 \# \text{ of characters in prompt}_i \\
 & + \beta_2 \text{professional terms}_i + \beta_3 \# \text{ of options}_i + \beta_4 \text{order in question}_i + \beta_5 \text{order in survey}_i \\
 & + \beta_6 \text{other}_i + \beta_7 \text{none}_i + \beta_8 \text{don't know}_i + \beta_9 \# \text{ of characters in prompt}_i * \text{professional terms}_i
 \end{aligned}$$

$$+ \beta_{10} \text{order in question}_i * \# \text{ of options}_i + \beta_{11} \text{other}_i * \# \text{ of characters in prompt}_i,$$

where i indicates each item/option/term

Equation 2. Logistic Regression Model with Interaction Terms from Correlation Analysis

To ensure the model's efficiency, the examination of variable importance is then tested using the “varImp” function in R. Table 3 showcases the results of this examination.

Independent Variable	Overall Importance
# of characters in prompt	85.3026
professional terms	1.5528
# of options	12.2271
order in question	24.2828
order in survey	8.4499
other	18.8243
none	58.3401
don't know	105.2920
interaction term of # of characters in prompt & professional terms	27.2424
interaction term of # of options & order in question	16.6487
interaction term of # of characters in prompt & other	13.3810

Table 3. Results of Variable Importance Analysis

The results suggest removing independent variables having importance scores of less than 10 from Equation 2, which are “parentheses and professional terms included” and “sequence of the question where the checkbox locates in the survey.” However, the reduced model with 6 main effects and 3 interaction terms does not achieve a smaller AIC score compared to the full model.

To determine which model yields the lowest AIC score, we utilize an iterative method known as the stepwise approach. The analysis reveals that retaining all predictors in Equation 2 results in the smallest, and thus most optimal, AIC score of 533122. Table 4 provides a summary of the stepwise approach results.

Change from the Full Model	Degree(s) of Freedom	Deviance	AIC
none		533098	533122
- order in survey	1	533170	533192
- interaction term of # of characters in prompt & other	1	533262	533284
- interaction term of # of options & order in question	1	533370	533392
- interaction term of # of characters in prompt & professional terms	1	533824	533846
- none	1	537874	537896
- don't know	1	548323	548345

Table 4. AIC Scores for Models Examined in Stepwise Approach

Based on the AIC scores, we move forward using Equation 2 where all 8 main effects and 3 interaction terms are incorporated.

Model Fitting and Evaluation of Fitness

After substituting the coefficients, the equation is represented as Equation 3. Coefficients represent the log odds of the checkbox being marked, given the values of the predictors. These coefficients can be interpreted as “the change in the log odds of the outcome for a one-unit change in the corresponding independent variable, holding all other variables constant.” Additional explanations regarding odds ratios and probabilities will be provided in the Results section.

$$\text{logit}(\text{Pr}(\text{possibility of marked}_i = 1)) = -0.1261 - 0.0171 \# \text{ of characters in prompt}_i$$

$$\begin{aligned}
& - 0.0475 \text{ professional terms}_{i,1} - 0.0124 \# \text{ of options}_i - 0.0656 \text{ order in question}_i \\
& + 0.0031 \text{ order in survey}_i - 3.3232 \text{ other}_{i,1} - 1.5695 \text{ none}_{i,1} - 2.0790 \text{ don't know}_{i,1} \\
& + 0.0115 \# \text{ of characters in prompt}_i * \text{ professional terms}_{i,1} \\
& + 0.0023 \# \text{ of options}_i * \text{ order in question}_i + 0.4624 \# \text{ of characters in prompt}_i * \text{ other}_{i,1},
\end{aligned}$$

where i indicates each item/option/term

Equation 3. Logistic Regression Equation with Coefficient Substitution

Several methods are applied to evaluate Equation 3. Firstly, the model obtains an AIC score of 533122 and a BIC score of 533256, both of which are the lowest among all models tested. Secondly, the generalized variance inflation factor (GVIF) reveals multicollinearity greater than 10 for several predictors, indicating a need for removal. However, further examination through ROC curve analysis indicates that both models achieve the same area under the curve value of 0.6466. Thus, we keep Equation 3 as the best-fitted model as it not only reaches the lowest AIC and BIC scores but also directly responds to our hypotheses.

Variable Name	GVIF
# of characters in prompt	2.3814
professional terms	9.9975
# of options	2.7432
order in question	11.5809
order in survey	1.2975
other	121.3246
none	1.0579
don't know	1.2186
interaction term of # of characters in prompt & professional terms	12.0362
interaction term of # of options & order in question	15.1946

interaction term of # of characters in prompt & other	120.5339
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Table 5. GVIF Scores for the Model in Equation 3

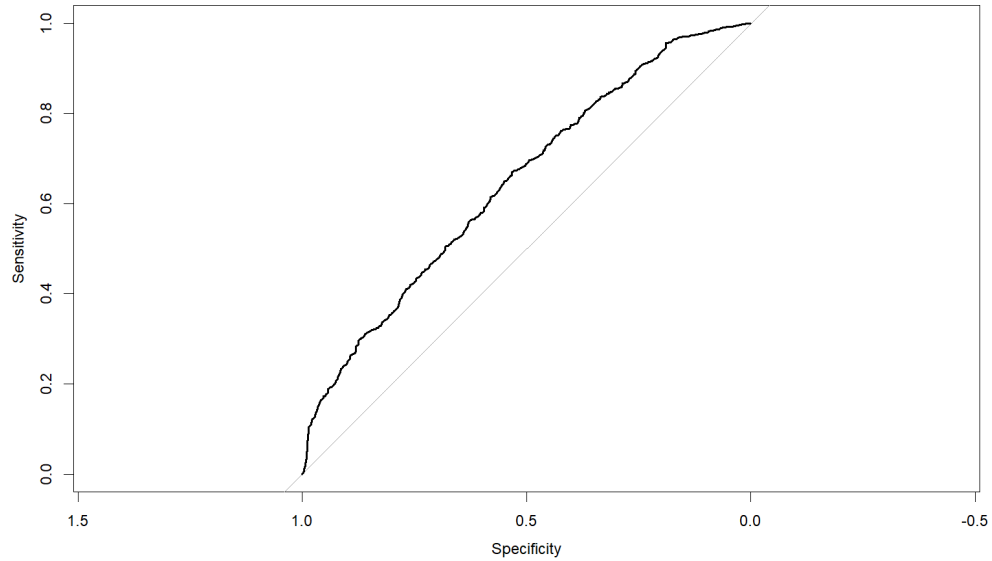


Figure 2. ROC Curve for the Model in Equation 3

Results

This section seeks to present the interpretations of the model coefficients in Equation 3. Table 6 displays the values of the estimated coefficients, their exponentiated values as odds or odds ratios, the probabilities associated with each coefficient estimates, and their statistical significance at a 95% confidence level placed right next to the coefficients.

Variable Name	Coefficient (Statistical Significance)	Odds of marked over not marked (e^{β})	Probability of marked ($\frac{e^{\beta}}{1+e^{\beta}}$)
Intercept	-0.1261***	0.8816	0.4685
# of characters in prompt	-0.0171***	0.9830	0.4957
professional terms	-0.0475	0.9536	0.4881
# of options	-0.0124***	0.9877	0.4969

order in question	-0.0656***	0.9365	0.4836
order in survey	0.0031***	1.0031	0.5008
other	-3.3231***	0.0360	0.0348
none	-1.5695***	0.2082	0.1723
don't know	-2.0790***	0.1251	0.1112
interaction term of # of characters in prompt & professional terms	0.0115***	1.0116	0.5029
interaction term of # of options & order in question	0.0023***	1.0023	0.5006
interaction term of # of characters in prompt & other	0.4624***	1.5879	0.6136

Table 6. Coefficient Estimates, Associated Odds/Odds Ratio, Associated Probability, and Statistical Significance at 95% Confidence Level

Based on Table 6 containing exponentiated coefficient estimates and the probabilities from the best-fitted logistic regression model, the interpretations are:

Main Effects (holding all other predictors constant; all but “professional terms” are statistically significant at 95% confidence level)

- When all other predictor variables are zero, the odds of the checkbox being endorsed is 0.8816, which is associated with the mean probability of 46.85%. The mean probability is statistically significantly different from 0 at the specified confidence level.
- Any additional character in the alternative is prone to decreased odds of the checkbox selection by 1.7%, which is calculated by

$$1 - e^{\# \text{ of characters in prompt}} = 1 - 0.9830 = 0.017$$
In other words, every one more character in the alternative prompt is associated with a lower probability of endorsement.
- Compared to alternatives that do not contain supplemental explanations or examples in parentheses, the odds of each checkbox selection decrease by 4.64% ($1 - e^{\text{professional terms}} = 1 - 0.9536 = 0.0464$) among options that include parentheses as supplemental examples or clarifications. However, the difference is not significant at our specified level of significance.

- Any additional alternative added to the single-list check-all-that-apply question is associated with decreased odds of checkbox selection. The magnitude of the decrease is 1.23% ($1 - e^{\# \text{ of options}} = 1 - 0.9877 = 0.0123$).
- Any one shift backward of the alternative within the question tends to decrease the odds of its selection by 6.35% ($1 - e^{\text{order in question}} = 1 - 0.9365 = 0.0635$).
- Any one shift backward of the question the alternative is located in within the survey tends to increase the odds of its endorsement by 0.31% ($e^{\text{order in survey}} - 1 = 1.0031 - 1 = 0.0031$).
- When compared to alternatives other than “Other,” the odds of endorsing “Other” over not endorsing significantly decrease by 96.4% ($1 - e^{\text{other}} = 1 - 0.0360 = 0.9640$).
- When compared to alternatives other than “None,” the odds of selecting “None” over not selecting decrease by 79.18% ($1 - e^{\text{none}} = 1 - 0.2082 = 0.7918$).
- When compared to alternatives other than “don’t know,” the odds of selecting “don’t know” over not selecting decrease by 87.49% ($1 - e^{\text{don't know}} = 1 - 0.1251 = 0.8749$).

Interaction Terms (holding all other predictors constant; all are statistically significant at 95% confidence level)

- The positive coefficient estimate for the interaction term “number of characters contained in an alternative” with “the existence of parentheses for explanations” indicates that the combined effect of these two variables increases the probability of the checkbox selection, compared to either of the main factors themselves while others remain the same.
- The positive coefficient for the interaction term “number of alternatives within a question” with “the location of the alternative within the same question” indicates that the combined effect of these two variables increases the probability of the checkbox selection, compared to either of the main factors themselves while the others remain the same.
- The positive coefficient for the interaction term “the number of characters contained in an alternative” with the alternative being “Other” or not indicates that the combined effect of these two variables increases the probability of the checkbox selection, compared to either of the main factors themselves while others remain the same.

Conclusions

Reviewing the hypotheses set forth at the beginning of this paper, it’s notable that two of the five hypotheses, specifically the second and last, are contradicted. The analysis indicates that the primacy effects stay true for options within the same survey item but not for different survey items within a survey wave. In other words, options listed earlier in an item gain more endorsement, while items located later in the survey gain more endorsement. Additionally,

Michigan local government officials demonstrate a tendency to refrain from selecting “None” or “Don’t Know” when presented, with both decreases in odds being tremendous at over 70%.

While mirroring public response patterns overall, the disparities noted in hypothesis testing outcomes indicate the necessity for tailored survey designs and outreach strategies for Michigan local government elites. Specifically, the findings alleviate concerns regarding randomizing survey items to mitigate primacy effects and the exclusion of “None” and “Don’t Know” options to avoid satisficing. Thus, based on the results, future research with a similar sampling frame should not be worried too seriously about the primacy effects on ordering survey items and the satisficing issues of presenting “None” and “Don’t Know” in the phase of survey design.

Limitations and Discussions

This study faces several limitations that hinder detailed examinations and comparison analyses. Firstly, complete waves of response data containing more sophisticated grid questions are inaccessible. Secondly, each wave includes approximately 2700 respondents. Invitations are extended to all officials on the sampling frame, without implementing a specific sampling approach. Thirdly, there is a lack of resources to compare officials and non-officials directly, which could have contributed significantly to the designs of future studies targeting the two population groups. Finally, cognitive interviews, paradata, and experiments were not included in this study’s design due to the constraints related to identifying and contacting appropriate respondents, but they could be beneficial in improving data quality from a qualitative perspective through direct feedback from the sample people (Nicolaas et al., 2015).

Future research is encouraged to explore further based on this study in the following directions:

- Explore differentiated response patterns stratified by jurisdiction type, providing valuable insights for survey design, strategic promotional approaches, and contact materials.
- For surveys conducted biannually or seasonally, analyze participation and response variations across different seasons to assess the potential impact of seasonal events, such as deer hunting season or elections, on data quality.
- Expand the study to encompass multiple states, recruiting a larger sampling frame to enhance the statistical power of analytic results and implement varied sampling strategies.
- Investigate contrasting responses of officials versus non-officials to identify distinct strategies, shedding light on the factors influencing survey participation and response behaviors among different demographic groups.
- Follow up with respondents to obtain direct insights into their responses through cognitive interviews or experiments.

These investigations can guide customized outreach strategies and refine survey designs to enhance engagement and data quality across various respondent demographics, particularly addressing the existing gap in research concerning local political officials.

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Appendix

A. Data Manipulation R codes

```
# Setup
library(haven)
library(dplyr)
library(tidyr)
library(ggplot2)
library(ggcorrplot)

# Load Data Sets
mpps_2009 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/mpps_data/2009spring_data_final_20130128.dta")
str(mpps_2009)
mpps_2010 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/2010spring_data_final_resopndents-V2-20151008.dta")
str(mpps_2010)
mpps_2011 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/2011spring_data_final_respondents_20151007.dta")
str(mpps_2011)
mpps_2012_fall <- read_dta("G:/My Drive/0. study
abroad/academic/11. 2024 Winter/3. SurvMeth 721 Total Survey
Error and Quality II/2. Assignments/MPPS
Data/2012fall_data_final_respondents-20160322.dta")
str(mpps_2012_fall)
mpps_2012_spring <- read_dta("G:/My Drive/0. study
abroad/academic/11. 2024 Winter/3. SurvMeth 721 Total Survey
Error and Quality II/2. Assignments/MPPS
Data/2012spring_data_final_respondents_20151007.dta")
```

```

str(mpps_2012_spring)
mpps_2013 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/2013spring_data_final_respondents-20150930.dta")
str(mpps_2013)
mpps_2014 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/2014spring_data_final_respondents-20150930.dta")
str(mpps_2014)
mpps_2015 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/2015spring_data_final_respondents-20151123.dta")
str(mpps_2015)
mpps_2016_fall <- read_dta("G:/My Drive/0. study
abroad/academic/11. 2024 Winter/3. SurvMeth 721 Total Survey
Error and Quality II/2. Assignments/MPPS
Data/2016fall_data_final_respondents_20161214.dta")
str(mpps_2016_fall)
mpps_2016_spring <- read_dta("G:/My Drive/0. study
abroad/academic/11. 2024 Winter/3. SurvMeth 721 Total Survey
Error and Quality II/2. Assignments/MPPS
Data/2016spring_data_final_respoondents-20160706.dta")
str(mpps_2016_spring)
mpps_2017 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/mpps_data/2017spring_data_final_respondents_20171023.dta")
str(mpps_2017)
mpps_2018 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/mpps_data/2018spring_data_final_respondents_20181011.dta")
str(mpps_2018)
mpps_2019 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/2019spring_data_final_respondents.dta")
str(mpps_2019)
mpps_2020 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS Data/2020spring_data_all_20200609.dta")
str(mpps_2020)

```

```

mpps_2021 <- read_dta("G:/My Drive/0. study abroad/academic/11.
2024 Winter/3. SurvMeth 721 Total Survey Error and Quality
II/2. Assignments/MPPS
Data/2021spring_data_final_20210726.dta")
str(mpps_2021)

# Exploratory Data Analysis
## 2009 spring
# see how many NAs are in gender responses in completed
responses
mpps_2009_com <- mpps_2009 %>%
  filter(idcompleted != 0) # 1455 obs
sum(is.na(mpps_2009_com$q33)) # 53 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2009_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2010 spring
# see how many NAs are in gender responses in completed
responses
mpps_2010_com <- mpps_2010 %>%
  filter(idcompleted != 0) # 1550 obs
sum(is.na(mpps_2010_com$q43)) # 64 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2010_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2011 spring
# see how many NAs are in gender responses in completed
responses
mpps_2011_com <- mpps_2011 %>%
  filter(idcompleted != 0) # 1340 obs
sum(is.na(mpps_2011_com$q62)) # 41 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2011_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2012 fall
# see how many NAs are in gender responses in completed
responses
mpps_2012_fall_com <- mpps_2012_fall %>%
  filter(idcompleted != 0) # 1409 obs

```

```

sum(is.na(mpps_2012_fall_com$q31)) # 45 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2012_fall_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2012 spring
# see how many NAs are in gender responses in completed
responses
mpps_2012_spring_com <- mpps_2012_spring %>%
  filter(idcompleted != 0) # 1398 obs
sum(is.na(mpps_2012_spring_com$q55)) # 38 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2012_spring_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2013 spring
# see how many NAs are in gender responses in completed
responses
mpps_2013_com <- mpps_2013 %>%
  filter(idcompleted != 0) # 1400 obs
sum(is.na(mpps_2013_com$q61)) # 73 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2013_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2014 spring
# see how many NAs are in gender responses in completed
responses
mpps_2014_com <- mpps_2014 %>%
  filter(idcompleted != 0) # 1417 obs
sum(is.na(mpps_2014_com$q42)) # 79 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2014_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2015 spring
# see how many NAs are in gender responses in completed
responses
mpps_2015_com <- mpps_2015 %>%
  filter(idcompleted != 0) # 1355 obs
sum(is.na(mpps_2015_com$q45)) # 85 obs

```

```

## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2015_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2016 fall
# see how many NAs are in gender responses in completed
responses
mpps_2016_fall_com <- mpps_2016_fall %>%
  filter(idcompleted != 0) # 1292 obs
sum(is.na(mpps_2016_fall_com$q28)) # 71 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2016_fall_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2016 spring
# see how many NAs are in gender responses in completed
responses
mpps_2016_spring_com <- mpps_2016_spring %>%
  filter(idcompleted != 0) # 1394 obs
sum(is.na(mpps_2016_spring_com$q42)) # 95 obs
## not considering comparing between two gender groups because
it'll reduce effective size of responses
sum(is.na(mpps_2016_spring_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2017 spring
mpps_2017_com <- mpps_2017 %>%
  filter(idcompleted != 0) # 1783 obs
sum(is.na(mpps_2017_com$q39)) # 125 obs
sum(is.na(mpps_2017_com$respondent_id_all)) # 0 obs
## all completed responses are assigned an id

## 2018 spring
mpps_2018_com <- mpps_2018 %>%
  filter(idcompleted != 0) # 1384 obs
sum(is.na(mpps_2018_com$q45)) # 101 obs
sum(is.na(mpps_2018_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2019 spring
mpps_2019_com <- mpps_2019 %>%
  filter(idcompleted != 0) # 1361 obs
sum(is.na(mpps_2019_com$q45)) # 99 obs

```

```

sum(is.na(mpps_2019_com$respondent_id)) # 0 obs
## all completed responses are assigned an id

## 2020 spring
mpps_2020_com <- mpps_2020 %>%
  filter(idcompleted != 0) # 1737 obs
sum(is.na(mpps_2020_com$q54)) # 141 obs
sum(is.na(mpps_2020_com$respondent_id_all)) # 0 obs
## all completed responses are assigned an id

## 2021 spring
mpps_2021_com <- mpps_2021 %>%
  filter(idcompleted != 0) # 1352 obs
sum(is.na(mpps_2021_com$q53)) # 122 obs
sum(is.na(mpps_2021_com$respondent_id)) # 0 obs
## all completed responses are assigned an id
which(names(mpps_2021_com) == 'respondent_id')
which(names(mpps_2021_com) == 'source')

# Data Cleaning/Data Preprocessing
## 2009 spring
# select required variables from completed cases
# 7, 10, 11, 15, 21
mpps_2009_res <- mpps_2009 %>%
  filter(idcompleted != 0) %>%
  select(1:8, 60:69, 97:130, 139:149, 165:170) %>%
  rename_at(vars(starts_with("q")), ~paste("2009_", ., sep =
""))
# long to wide data set
mpps_2009_res_wide <- mpps_2009_res %>%
  select(1, 3, 9:69) %>%
  gather(key = "question", value = response, -respondent_id,
-source)

## 2010 spring
# select required variables from completed cases
# 8, 10, 14, 24, 28, 31
colnames(mpps_2010)
mpps_2010_res <- mpps_2010 %>%
  filter(idcompleted != 0) %>%
  select(1:8, 63:66, 72:75, 86:89, 100:107, 111:122, 125:136)
%>%
  rename_at(vars(starts_with("q")), ~paste("2010_", ., sep =
""))
# long to wide data set

```

```

mpps_2010_res_wide <- mpps_2010_res %>%
  select(1, 3, 9:52) %>%
  gather(key = "question", value = response, -respondent_id,
    -source)

## 2011 spring
# select required variables from completed cases
# 14, 16, 20, 24, 35, 42
colnames(mpps_2011)
mpps_2011_res <- mpps_2011 %>%
  filter(idcompleted != 0) %>%
  select(1:8, 79:82, 87:90, 101:104, 108:112, 129:134, 143:153)
%>%
  rename_at(vars(starts_with("q")), ~paste("2011_", ., sep =
    ""))
# long to wide data set
mpps_2011_res_wide <- mpps_2011_res %>%
  select(1, 3, 9:42) %>%
  gather(key = "question", value = response, -respondent_id,
    -source)

## 2012 fall
# select required variables from completed cases
# 4, 10, 22, 28, 29
colnames(mpps_2012_fall)
mpps_2012_fall_res <- mpps_2012_fall %>%
  filter(idcompleted != 0) %>%
  select(1:8, 23:42, 76:79, 110:118, 128:151) %>%
  rename_at(vars(starts_with("q")), ~paste("2012_f_", ., sep =
    ""))
# long to wide data set
mpps_2012_fall_res_wide <- mpps_2012_fall_res %>%
  select(1, 3, 9:65) %>%
  gather(key = "question", value = response, -respondent_id,
    -source)

## 2012 spring
# select required variables from completed cases
# 8, 22, 24
colnames(mpps_2012_spring)
mpps_2012_spring_res <- mpps_2012_spring %>%
  filter(idcompleted != 0) %>%
  select(1:8, 69:74, 121:124, 131:134) %>%
  rename_at(vars(starts_with("q")), ~paste("2012_s_", ., sep =
    ""))

```



```

# long to wide data set
mpps_2012_spring_res_wide <- mpps_2012_spring_res %>%
  select(1, 3, 9:22) %>%
  gather(key = "question", value = response, -respondent_id,
    -source)

## 2013 spring
# select required variables from completed cases
# 8, 16, 18 (grid), 31, 37, 43
colnames(mpps_2013)
mpps_2013_res <- mpps_2013 %>%
  filter(idcompleted != 0) %>%
  select(1:8, 69:74, 86:90, 137:140, 150:163, 174:188) %>%
  rename_at(vars(starts_with("q")), ~paste("2013_", ., sep =
    ""))
# long to wide data set
mpps_2013_res_wide <- mpps_2013_res %>%
  select(1, 3, 9:52) %>%
  gather(key = "question", value = response, -respondent_id,
    -source)

## 2014 spring
# select required variables from completed cases
# 11, 14, 16, 18, 19
colnames(mpps_2014)
mpps_2014_res <- mpps_2014 %>%
  filter(idcompleted != 0) %>%
  select(1:8, 76:101, 108:113, 115:123, 126:157) %>%
  rename_at(vars(starts_with("q")), ~paste("2014_", ., sep =
    ""))
# long to wide data set
mpps_2014_res_wide <- mpps_2014_res %>%
  select(1, 3, 9:81) %>%
  gather(key = "question", value = response, -respondent_id,
    -source)

## 2015 spring
# select required variables from completed cases
# 17 (grid), 19, 29
colnames(mpps_2015)
mpps_2015_res <- mpps_2015 %>%
  filter(idcompleted != 0) %>%
  select(1:8, 117:129, 148:161) %>%
  rename_at(vars(starts_with("q")), ~paste("2015_", ., sep =
    ""))

```

```

# long to wide data set
mpps_2015_res_wide <- mpps_2015_res %>%
  select(1, 3, 9:35) %>%
  gather(key = "question", value = response, -respondent_id,
    -source)

## 2016 fall
# select required variables from completed cases
# no check-all-that-apply question other than the demographic
(race)

## 2016 spring
# select required variables from completed cases
# 30
colnames(mpps_2016_spring)
mpps_2016_spring_res <- mpps_2016_spring %>%
  filter(idcompleted != 0) %>%
  select(1:8, 160:163) %>%
  rename_at(vars(starts_with("q")), ~paste("2016_", ., sep =
    ""))
# long to wide data set
mpps_2016_spring_res_wide <- mpps_2016_spring_res %>%
  select(1, 3, 9:12) %>%
  gather(key = "question", value = response, -respondent_id,
    -source)
sum(is.na(mpps_2016_spring_res_wide$response)) # 3356 NAs

## 2017 spring
# select required variables from completed cases
# 18, 23, 26
mpps_2017_res <- mpps_2017 %>%
  filter(idcompleted != 0) %>%
  select(1:10, 98:103, 118:124, 127:134) %>%
  rename_at(vars(starts_with("q")), ~paste("2017_", ., sep =
    ""))
# long to wide data set
colnames(mpps_2017_res)
mpps_2017_res_wide <- mpps_2017_res %>%
  select(3, 5, 11:31) %>%
  gather(key = "question", value = response,
    -respondent_id_all, -source) %>%
  rename(respondent_id = respondent_id_all)

## 2018 spring

```

```

# select required variables from completed cases
# 22, 29, 31
mpps_2018_res <- mpps_2018 %>%
  filter(idcompleted != 0) %>%
  select(1:8, 94:101, 110:118, 120:124) %>%
  rename_at(vars(starts_with("q")), ~paste("2018_", ., sep =
""))
# long to wide data set
mpps_2018_res_wide <- mpps_2018_res %>%
  select(1, 3, 9:30) %>%
  gather(key = "question", value = response, -respondent_id,
-source)

## 2019 spring
# select required variables from completed cases
# 23, 27, 28, 30, 33, 34
colnames(mpps_2019)
mpps_2019_res <- mpps_2019 %>%
  filter(idcompleted != 0) %>%
  select(1:8, 88:97, 105:123, 126:136, 145:163) %>%
  rename_at(vars(starts_with("q")), ~paste("2019_", ., sep =
""))
# long to wide data set
mpps_2019_res_wide <- mpps_2019_res %>%
  select(1, 3, 9:67) %>%
  gather(key = "question", value = response, -respondent_id,
-source)

## 2020 spring
# select required variables from completed cases
# 26, 33, 39
colnames(mpps_2020)
mpps_2020_res <- mpps_2020 %>%
  filter(idcompleted != 0) %>%
  select(1:10, 117:126, 144:152, 163:173) %>%
  rename_at(vars(starts_with("q")), ~paste("2020_", ., sep =
""))
# long to wide data set
colnames(mpps_2020_res)
mpps_2020_res_wide <- mpps_2020_res %>%
  select(3, 5, 11:40) %>%
  gather(key = "question", value = response,
-respondent_id_all, -source)%>%
  rename(respondent_id = respondent_id_all)

```



```

    nu_option = c(9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 15, 15, 15, 15, 15,
15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 15, 17, 17, 17, 17, 17, 17,
17, 17, 17, 17, 17, 17, 17, 17, 17, 17, 17, 10, 10, 10, 10, 10,
10, 10, 10, 10, 10, 6, 6, 6, 6, 6, 6),
    order_prompt = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 2, 3, 4, 5, 6,
7, 8, 9, 10, 11, 12, 13, 14, 15, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10,
11, 12, 13, 14, 15, 16, 17, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 1,
2, 3, 4, 5, 6),
    order_survey = c(7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 10, 10, 10, 10,
10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 11, 11, 11, 11, 11,
11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 15, 15, 15, 15,
15, 15, 15, 15, 15, 15, 21, 21, 21, 21, 21, 21),
    other = c(0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0),
    none = c(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0),
    dk = c(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1)
)
# apply the mappings using case_when
mpps_2009_res_wide_digit <- mpps_2009_res_wide %>%
  filter(!(question %in% c("2009_q7j", "2009_q10p",
"2009_q11r", "2009_q15k"))) %>%
  left_join(mapping_table_2009, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

## 2010 spring
mapping_table_2010 <- tibble(
  question = c("2010_q8a", "2010_q8b", "2010_q8c", "2010_q8d",
"2010_q10a", "2010_q10b", "2010_q10c", "2010_q10d",
"2010_q14a", "2010_q14b", "2010_q14c", "2010_q14d",
"2010_q24a", "2010_q24b", "2010_q24c", "2010_q24d",
"2010_q24e", "2010_q24f", "2010_q24g", "2010_q28a",
"2010_q28b", "2010_q28c", "2010_q28d", "2010_q28e",
"2010_q28f", "2010_q28g", "2010_q28h", "2010_q28i",
"2010_q28j", "2010_q28k", "2010_q31a", "2010_q31b",
"2010_q31c", "2010_q31d", "2010_q31e", "2010_q31f",
"2010_q31g", "2010_q31h", "2010_q31i", "2010_q31j",
"2010_q31k"),
  nu_character_prompt = c(58, 64, 2, 10, 64, 73, 2, 10, 41, 36,
2, 10, 3, 3, 11, 15, 11, 5, 10, 72, 83, 36, 35, 52, 61, 29, 5,
32, 49, 10, 105, 94, 85, 133, 78, 71, 35, 57, 56, 5, 10),

```



```

8, 8, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 12, 12, 12, 12,
12, 12, 12, 12, 12, 12, 12, 12),
  order_prompt = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,
14, 15, 16, 17, 18, 19, 20, 1, 2, 3, 4, 1, 2, 3, 4, 5, 6, 7, 8,
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10,
11, 12),
  order_survey = c(4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,
4, 4, 4, 4, 4, 10, 10, 10, 10, 22, 22, 22, 22, 22, 22, 22, 22,
28, 28, 28, 28, 28, 28, 28, 28, 28, 28, 28, 28, 29, 29, 29, 29, 29, 29,
29, 29, 29, 29, 29, 29),
  other = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0),
  none = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
  dk = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)
)
# apply the mappings using case_when
mpps_2012_fall_res_wide_digit <- mpps_2012_fall_res_wide %>%
  filter(!(question %in% c("2012_f_q22i", "2012_f_q28k",
"2012_f_q29m"))) %>%
  left_join(mapping_table_2012_fall, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

## 2012 spring
mapping_table_2012_spring <- tibble(
  question = c("2012_s_q8a", "2012_s_q8b", "2012_s_q8c",
"2012_s_q8d", "2012_s_q8e", "2012_s_q8f", "2012_s_q22a",
"2012_s_q22b", "2012_s_q22c", "2012_s_q22d", "2012_s_q24a",
"2012_s_q24b", "2012_s_q24c", "2012_s_q24d"),
  nu_character_prompt = c(18, 23, 75, 111, 11, 10, 58, 64, 2,
10, 64, 73, 2, 10),
  professional_terms = c(0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0),
  nu_option = c(6, 6, 6, 6, 6, 6, 4, 4, 4, 4, 4, 4, 4, 4),
  order_prompt = c(1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 1, 2, 3, 4),
  order_survey = c(8, 8, 8, 8, 8, 8, 22, 22, 22, 22, 24, 24,
24, 24),
  other = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
  none = c(0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0),
  dk = c(0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1)
)

```



```

# apply the mappings using case_when
mpps_2012_spring_res_wide_digit <- mpps_2012_spring_res_wide
%>%
  left_join(mapping_table_2012_spring, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

## 2013 spring
mapping_table_2013 <- tibble(
  question = c("2013_q8a", "2013_q8b", "2013_q8c", "2013_q8d",
"2013_q8e", "2013_q8f", "2013_q16a", "2013_q16b", "2013_q16c",
"2013_q16d", "2013_q16e", "2013_q31a", "2013_q31b",
"2013_q31c", "2013_q31d", "2013_q37a", "2013_q37b",
"2013_q37c", "2013_q37d", "2013_q37e", "2013_q37f",
"2013_q37g", "2013_q37h", "2013_q37i", "2013_q37j",
"2013_q37k", "2013_q37l", "2013_q37m", "2013_q43a",
"2013_q43b", "2013_q43c", "2013_q43d", "2013_q43e",
"2013_q43f", "2013_q43g", "2013_q43h", "2013_q43i",
"2013_q43j", "2013_q43k", "2013_q43l", "2013_q43m",
"2013_q43n"),
  nu_character_prompt = c(18, 23, 75, 111, 11, 10, 6, 4, 28, 8,
10, 58, 64, 2, 10, 17, 15, 48, 74, 29, 46, 41, 62, 34, 28, 46,
5, 10, 37, 99, 58, 63, 26, 82, 32, 57, 36, 45, 60, 135, 5, 10),
  professional_terms = c(0, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0,
0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1,
0, 1, 0, 1, 1, 1, 0, 0),
  nu_option = c(6, 6, 6, 6, 6, 6, 5, 5, 5, 5, 5, 4, 4, 4, 4,
13, 13, 13, 13, 13, 13, 13, 13, 13, 13, 13, 13, 13, 13, 14, 14, 14,
14, 14, 14, 14, 14, 14, 14, 14, 14, 14, 14),
  order_prompt = c(1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 1, 2, 3, 4,
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 1, 2, 3, 4, 5, 6, 7,
8, 9, 10, 11, 12, 13, 14),
  order_survey = c(8, 8, 8, 8, 8, 8, 16, 16, 16, 16, 16, 31,
31, 31, 31, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37, 37,
43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43, 43),
  other = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 1, 0),
  none = c(0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0),
  dk = c(0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 1)
)
# apply the mappings using case_when

```

```

mpps_2013_res_wide_digit <- mpps_2013_res_wide %>%
  filter(!(question %in% c("2013_q37n", "2013_q43o"))) %>%
  left_join(mapping_table_2013, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

# 2014 spring
mapping_table_2014 <- tibble(
  question = c("2014_q11a", "2014_q11b", "2014_q11c",
"2014_q11d", "2014_q11e", "2014_q11f", "2014_q11g",
"2014_q11h", "2014_q11i", "2014_q11j", "2014_q11k",
"2014_q11l", "2014_q11m", "2014_q11n", "2014_q11o",
"2014_q11p", "2014_q11q", "2014_q11r", "2014_q11s",
"2014_q11t", "2014_q11u", "2014_q11v", "2014_q11w",
"2014_q11x", "2014_q11y", "2014_q14a", "2014_q14b",
"2014_q14c", "2014_q14d", "2014_q14e", "2014_q16a",
"2014_q16b", "2014_q16c", "2014_q16d", "2014_q16e",
"2014_q16f", "2014_q16g", "2014_q16h", "2014_q18a",
"2014_q18b", "2014_q18c", "2014_q18d", "2014_q18e",
"2014_q18f", "2014_q18g", "2014_q18h", "2014_q18i",
"2014_q18j", "2014_q18k", "2014_q18l", "2014_q18m",
"2014_q18n", "2014_q18o", "2014_q18p", "2014_q19a",
"2014_q19b", "2014_q19c", "2014_q19d", "2014_q19e",
"2014_q19f", "2014_q19g", "2014_q19h", "2014_q19i",
"2014_q19j", "2014_q19k", "2014_q19l", "2014_q19m", "q19n"),
  nu_character_prompt = c(21, 15, 48, 52, 18, 43, 28, 21, 31,
23, 15, 11, 9, 50, 17, 13, 34, 51, 27, 22, 27, 21, 53, 5, 10,
19, 4, 40, 5, 10, 32, 50, 30, 41, 33, 76, 5, 10, 37, 44, 50,
58, 34, 40, 51, 59, 22, 40, 48, 41, 32, 34, 5, 10, 43, 55, 40,
64, 22, 40, 48, 41, 32, 38, 39, 15, 5, 10),
  professional_terms = c(0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0,
1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
  nu_option = c(25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 25,
25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 25, 5, 5, 5, 5,
5, 8, 8, 8, 8, 8, 8, 8, 8, 16, 16, 16, 16, 16, 16, 16, 16, 16,
16, 16, 16, 16, 16, 16, 16, 14, 14, 14, 14, 14, 14, 14, 14, 14,
14, 14, 14, 14, 14),
  order_prompt = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,
14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 1, 2, 3, 4, 5,
1, 2, 3, 4, 5, 6, 7, 8, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12,
13, 14, 15, 16, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14),
  order_survey = c(11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11,
11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 11, 14, 14,
14, 14, 14, 16, 16, 16, 16, 16, 16, 16, 16, 18, 18, 18, 18, 18,

```

[illegible]

```

    dk = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 1)
)
# apply the mappings using case_when
mpps_2015_res_wide_digit <- mpps_2015_res_wide %>%
  filter(!(question %in% c("2015_q19other", "2015_q29other")))
%>%
  left_join(mapping_table_2015, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

# 2016 spring
mapping_table_2016_spring <- tibble(
  question = c("2016_q30a", "2016_q30b", "2016_q30c",
"2016_q30d"),
  nu_character_prompt = c(84, 86, 45, 10),
  professional_terms = c(0, 0, 0, 0),
  nu_option = c(4, 4, 4, 4),
  order_prompt = c(1, 2, 3, 4),
  order_survey = c(30, 30, 30, 30),
  other = c(0, 0, 0, 0),
  none = c(0, 0, 0, 0),
  dk = c(0, 0, 0, 1)
)
# apply the mappings using case_when
mpps_2016_spring_res_wide_digit <- mpps_2016_spring_res_wide
%>%
  left_join(mapping_table_2016_spring, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

## 2017 spring
# create a tibble with all information
mapping_table_2017 <- tibble(
  question = c("2017_q18a", "2017_q18b", "2017_q18c",
"2017_q18d", "2017_q18e", "2017_q18f", "2017_q23a",
"2017_q23b", "2017_q23c", "2017_q23d", "2017_q23e",
"2017_q23f", "2017_q26a", "2017_q26b", "2017_q26c",
"2017_q26d", "2017_q26e", "2017_q26f", "2017_q26g"),
  nu_character_prompt = c(49, 9, 9, 8, 19, 10, 29, 58, 70, 14,
5, 10, 32, 46, 39, 33, 36, 5, 10),
  professional_terms = c(0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0),
  nu_option = c(6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 7, 7, 7, 7,
7, 7, 7),
  order_prompt = c(1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 6, 1, 2, 3,
4, 5, 6, 7),

```

```

    order_survey = c(18, 18, 18, 18, 18, 18, 23, 23, 23, 23, 23,
23, 26, 26, 26, 26, 26, 26, 26),
    other = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
1, 0),
    none = c(1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0),
    dk = c(0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
1)
  )
# apply the mappings using case_when
mpps_2017_res_wide_digit <- mpps_2017_res_wide %>%
  filter(!(question %in% c("2017_q23other", "2017_q26other")))
%>%
  left_join(mapping_table_2017, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

## 2018 spring
# create a tibble with all information
mapping_table_2018 <- tibble(
  question = c("2018_q22a", "2018_q22b", "2018_q22c",
"2018_q22d", "2018_q22e", "2018_q22f", "2018_q22g",
"2018_q22h", "2018_q29a", "2018_q29b", "2018_q29c",
"2018_q29d", "2018_q29e", "2018_q29f", "2018_q29g",
"2018_q29h", "2018_q29i", "2018_q31a", "2018_q31b",
"2018_q31c", "2018_q31d", "2018_q31e"),
  nu_character_prompt = c(17, 79, 45, 44, 72, 35, 32, 21, 41,
60, 48, 31, 60, 68, 24, 37, 10, 61, 54, 45, 36, 10),
  professional_terms = c(0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0,
1, 0, 0, 0, 0, 0, 0, 0),
  nu_option = c(8, 8, 8, 8, 8, 8, 8, 8, 9, 9, 9, 9, 9, 9, 9, 9,
9, 5, 5, 5, 5, 5),
  order_prompt = c(1, 2, 3, 4, 5, 6, 7, 8, 1, 2, 3, 4, 5, 6, 7,
8, 9, 1, 2, 3, 4, 5),
  order_survey = c(22, 22, 22, 22, 22, 22, 22, 22, 29, 29, 29,
29, 29, 29, 29, 29, 29, 31, 31, 31, 31, 31),
  other = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0),
  none = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0),
  dk = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0,
0, 0, 0, 1)
)
# apply the mappings using case_when
mpps_2018_res_wide_digit <- mpps_2018_res_wide %>%
  left_join(mapping_table_2018, by = "question") %>%

```

```

mutate(across(starts_with("question_"), as.character))

## 2019 spring
# create a tibble with all information
mapping_table_2019 <- tibble(
  question = c("2019_q23a", "2019_q23b", "2019_q23c",
"2019_q23d", "2019_q23e", "2019_q23f", "2019_q23g",
"2019_q23h", "2019_q23i", "2019_q27a", "2019_q27b",
"2019_q27c", "2019_q27d", "2019_q27e", "2019_q27f",
"2019_q27g", "2019_q27h", "2019_q27i", "2019_q28a",
"2019_q28b", "2019_q28c", "2019_q28d", "2019_q28e",
"2019_q28f", "2019_q28g", "2019_q28h", "2019_q28i",
"2019_q30a", "2019_q30b", "2019_q30c", "2019_q30d",
"2019_q30e", "2019_q30f", "2019_q30g", "2019_q30h",
"2019_q30i", "2019_q30j", "2019_q33a", "2019_q33b",
"2019_q33c", "2019_q33d", "2019_q33e", "2019_q33f",
"2019_q33g", "2019_q33h", "2019_q33i", "2019_q34a",
"2019_q34b", "2019_q34c", "2019_q34d", "2019_q34e",
"2019_q34f", "2019_q34g", "2019_q34h", "2019_q34i"),
  nu_character_prompt = c(54, 31, 51, 68, 86, 77, 16, 5, 10,
60, 41, 35, 41, 27, 34, 43, 66, 10, 51, 55, 22, 82, 27, 60, 72,
5, 10, 51, 45, 36, 28, 63, 43, 41, 5, 14, 10, 43, 53, 132, 80,
67, 86, 65, 5, 10, 16, 25, 35, 81, 12, 24, 36, 108, 10),
  professional_terms = c(0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0),
  nu_option = c(9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9,
9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9,
10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10,
10, 10, 10, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9,
9),
  order_prompt = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 2, 3, 4, 5, 6,
7, 8, 9, 1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 2, 3, 4, 5, 6, 7, 8, 9,
10, 1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 2, 3, 4, 5, 6, 7, 8, 9),
  order_survey = c(23, 23, 23, 23, 23, 23, 23, 23, 23, 27, 27,
27, 27, 27, 27, 27, 27, 27, 28, 28, 28, 28, 28, 28, 28, 28, 28,
30, 30, 30, 30, 30, 30, 30, 30, 30, 30, 30, 33, 33, 33, 33, 33, 33,
33, 33, 33, 34, 34, 34, 34, 34, 34, 34, 34, 34, 34),
  other = c(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0,
0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
  none = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),

```

```

    dk = c(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1,
0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1)
)
# apply the mappings using case_when
mpps_2019_res_wide_digit <- mpps_2019_res_wide %>%
  filter(!(question %in% c("2019_q23other", "2019_q28other",
"2019_q30other", "2019_q33other"))) %>%
  left_join(mapping_table_2019, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

## 2020 spring
# create a tibble with all information
mapping_table_2020 <- tibble(
  question = c("2020_q26a", "2020_q26b", "2020_q26c",
"2020_q26d", "2020_q26e", "2020_q26f", "2020_q26g",
"2020_q26h", "2020_q26i", "2020_q33a", "2020_q33b",
"2020_q33c", "2020_q33d", "2020_q33e", "2020_q33f",
"2020_q33g", "2020_q33h", "2020_q39a", "2020_q39b",
"2020_q39c", "2020_q39d", "2020_q39e", "2020_q39f",
"2020_q39g", "2020_q39h", "2020_q39i", "2020_q39j"),
  nu_character_prompt = c(33, 43, 35, 33, 35, 54, 50, 5, 10,
33, 35, 33, 35, 54, 50, 5, 10, 41, 45, 49, 127, 99, 67, 86, 65,
5, 10),
  professional_terms = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0),
  nu_option = c(9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 8, 8, 8, 8, 8, 8, 8,
8, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10),
  order_prompt = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 2, 3, 4, 5, 6,
7, 8, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10),
  order_survey = c(26, 26, 26, 26, 26, 26, 26, 26, 26, 33, 33,
33, 33, 33, 33, 33, 33, 39, 39, 39, 39, 39, 39, 39, 39, 39,
39),
  other = c(0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0,
0, 0, 0, 0, 0, 0, 0, 1, 0),
  none = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0),
  dk = c(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0,
0, 0, 0, 0, 0, 0, 0, 1)
)
# apply the mappings using case_when
mpps_2020_res_wide_digit <- mpps_2020_res_wide %>%
  filter(!(question %in% c("2020_q26other", "2020_q33other",
"2020_q39other"))) %>%
  left_join(mapping_table_2020, by = "question") %>%

```

```

mutate(across(starts_with("question_"), as.character))

## 2021 spring
# create a tibble with all information
mapping_table_2021 <- tibble(
  question = c("2021_q17a", "2021_q17b", "2021_q17c",
"2021_q17d", "2021_q17e", "2021_q17f", "2021_q17g",
"2021_q17h", "2021_q17i", "2021_q18a", "2021_q18b",
"2021_q18c", "2021_q18d", "2021_q18e", "2021_q18f",
"2021_q27a", "2021_q27b", "2021_q27c", "2021_q27d",
"2021_q27e", "2021_q28a", "2021_q28b", "2021_q28c",
"2021_q28d", "2021_q28e", "2021_q28f", "2021_q28g",
"2021_q28h", "2021_q28i", "2021_q28j", "2021_q31a",
"2021_q31b", "2021_q31c", "2021_q31d"),
  nu_character_prompt = c(39, 23, 68, 48, 56, 88, 47, 5, 10,
43, 46, 41, 45, 5, 10, 23, 78, 71, 7, 10, 82, 88, 62, 107, 85,
69, 51, 77, 14, 10, 47, 26, 2, 10),
  professional_terms = c(0, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0,
0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0),
  nu_option = c(9, 9, 9, 9, 9, 9, 9, 9, 9, 6, 6, 6, 6, 6, 6, 5,
5, 5, 5, 5, 10, 10, 10, 10, 10, 10, 10, 10, 10, 4, 4, 4,
4),
  order_prompt = c(1, 2, 3, 4, 5, 6, 7, 8, 9, 1, 2, 3, 4, 5, 6,
1, 2, 3, 4, 5, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 1, 2, 3, 4),
  order_survey = c(17, 17, 17, 17, 17, 17, 17, 17, 17, 17, 18, 18,
18, 18, 18, 18, 27, 27, 27, 27, 27, 28, 28, 28, 28, 28, 28, 28,
28, 28, 28, 31, 31, 31, 31),
  other = c(0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0),
  none = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0),
  dk = c(0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0,
0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1)
)
# apply the mappings using case_when
mpps_2021_res_wide_digit <- mpps_2021_res_wide %>%
  filter(!(question %in% c("2021_q17other", "2021_q18other")))
%>%
  left_join(mapping_table_2021, by = "question") %>%
  mutate(across(starts_with("question_"), as.character))

## Combine data frames to one
mpps_2012_spring_res_wide_digit$response <-
as.character(mpps_2012_spring_res_wide_digit$response)

```



```

mpps_2016_spring_res_wide_digit$response <-
as.character(mpps_2016_spring_res_wide_digit$response)
mpps_2018_res_wide_digit$response <-
as.character(mpps_2018_res_wide_digit$response)

combined_list <- dplyr::bind_rows(mpps_2009_res_wide_digit,
mpps_2010_res_wide_digit, mpps_2011_res_wide_digit,
mpps_2012_fall_res_wide_digit, mpps_2012_spring_res_wide_digit,
mpps_2013_res_wide_digit, mpps_2014_res_wide_digit,
mpps_2015_res_wide_digit, mpps_2016_spring_res_wide_digit,
mpps_2017_res_wide_digit, mpps_2018_res_wide_digit,
mpps_2019_res_wide_digit, mpps_2020_res_wide_digit,
mpps_2021_res_wide_digit)

colnames(combined_list)
sum(is.na(combined_list$response))## 201332/713017 obs are NAs

table(mpps_2009_res_wide_digit$response)
table(mpps_2010_res_wide_digit$response)
table(mpps_2011_res_wide_digit$response)
table(mpps_2012_fall_res_wide_digit$response)
table(mpps_2012_spring_res_wide_digit$response)
table(mpps_2013_res_wide_digit$response)
table(mpps_2014_res_wide_digit$response)
table(mpps_2015_res_wide_digit$response)
table(mpps_2016_spring_res_wide_digit$response)
table(mpps_2017_res_wide_digit$response)
table(mpps_2018_res_wide_digit$response)
table(mpps_2019_res_wide_digit$response)
table(mpps_2020_res_wide_digit$response)
table(mpps_2021_res_wide_digit$response)
table(combined_list$response)

# Analytics
combined_list_excluded <- subset(combined_list, select =
-c(question, respondent_id, source))
combined_list_com <-
combined_list_excluded[complete.cases(combined_list_excluded),
]
# Descriptive Statistics
# Plot correlation matrix as a heatmap
library(ggplot2)
library(ggcorrplot)
## plotting the correlations between variables of interest
model.matrix(~0+., data = combined_list_excluded) %>%

```

```

cor(use = "pairwise.complete.obs", method = "spearman") %>%
ggcorrplot(show.diag = FALSE, type = "lower",
            lab = TRUE, lab_size = 5)
# Model Fitting
## take out source and year
list_logit_model_new <- glm(as.numeric(response) ~
nu_character_prompt + factor(professional_terms) + nu_option +
order_prompt + order_survey + factor(other) + factor(none) +
factor(dk) + nu_character_prompt:factor(professional_terms) +
order_prompt:nu_option + factor(other):nu_character_prompt,
data = combined_list, family = binomial, na.action =
na.exclude)
summary(list_logit_model_new) ## AIC: 527192
library(caret)
varImp(list_logit_model_new)
step_model <- step(list_logit_model_new)
summary(step_model)
list_logit_model_step <- glm(formula = as.numeric(response) ~
nu_character_prompt + professional_terms + nu_option +
order_prompt + order_survey + other + none + dk +
nu_character_prompt:professional_terms + nu_option:order_prompt
+ nu_character_prompt:other, family = binomial, data =
combined_list, na.action = na.exclude) ## AIC: 533122
summary(list_logit_model_step)
# Model Evaluation
# AIC and BIC
AIC(list_logit_model_step) ## 533122.3
BIC(list_logit_model_step) ## 533256

varImp(list_logit_model_step)

list_logit_model_step_varImp <- glm(formula =
as.numeric(response) ~ nu_character_prompt + nu_option +
order_prompt + other + none + dk + nu_option:order_prompt +
nu_character_prompt:other, family = binomial, data =
combined_list, na.action = na.exclude)
summary(list_logit_model_step_varImp)
AIC(list_logit_model_step_varImp)

# Check multicollinearity
library(car)
vif(list_logit_model_step, type = "predictor") ## other and
nu_character_prompt:other are the highest

# ROC curve analysis

```

```
library(pROC)
roc_curve <- roc(as.numeric(response) ~
predict(list_logit_model_step, type = "response"), data =
combined_list)
plot(roc_curve)

# AUC calculation
auc(roc_curve) ## 0.6466

# Residual plots
plot(list_logit_model_step, which = c(1, 2)) # Standardized
residuals vs. fitted values
plot(list_logit_model_step, which = 3)      # QQ plot of
residuals

# Influence diagnostics
influenceIndexPlot(list_logit_model_step) # Influence plot
```